



Cohort canonical evaluation — 10 agents

idea-01-lateral-attribution · 130 held-out segments · V0
yaw 0.01456 rad/s, CTE 147.44 m · generated
2026-05-31 21:46

- Executive summary

AGENTS (OK / TOTAL)

9 / 10

EVAL POOL

130 segs

V0 YAW RMSE

0.01456

V0 CTE RMSE

147.44 m

WALL TIME

3.26s

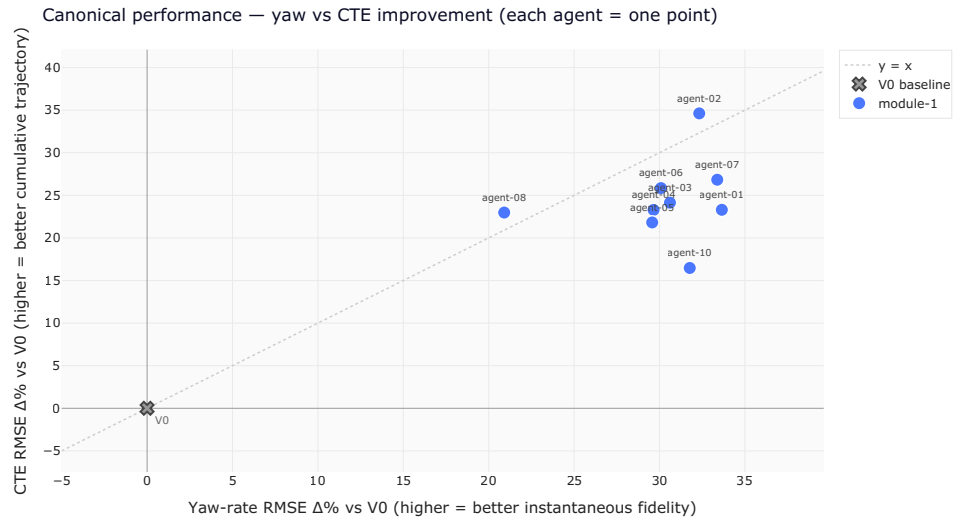
Best yaw: m1-agent-01 **+33.6%** · Best CTE: m1-agent-02

+34.6%

Winning both KPIs $\geq +30\%$ (1 agents): m1-agent-02

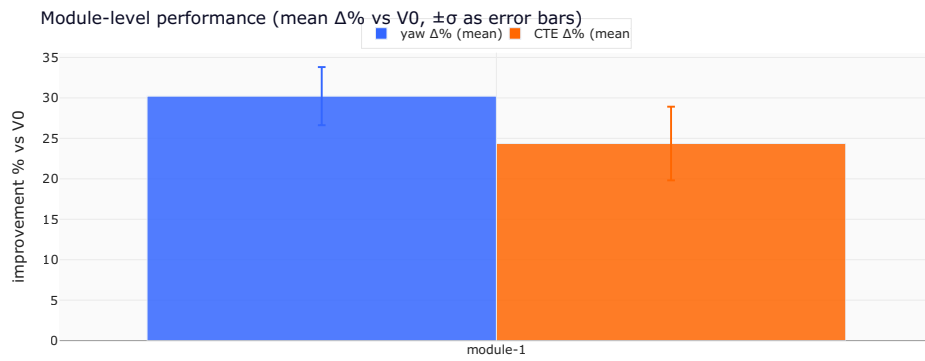
1. Headline scatter — yaw vs CTE improvement

Each agent is one point. **Top-right is the goal** — both KPIs improve over V0. The dashed diagonal is $y = x$: above it, an agent improved CTE more than yaw (good trajectory integration); below, the opposite. The V0 baseline sits at the origin.



2. Module-level aggregation

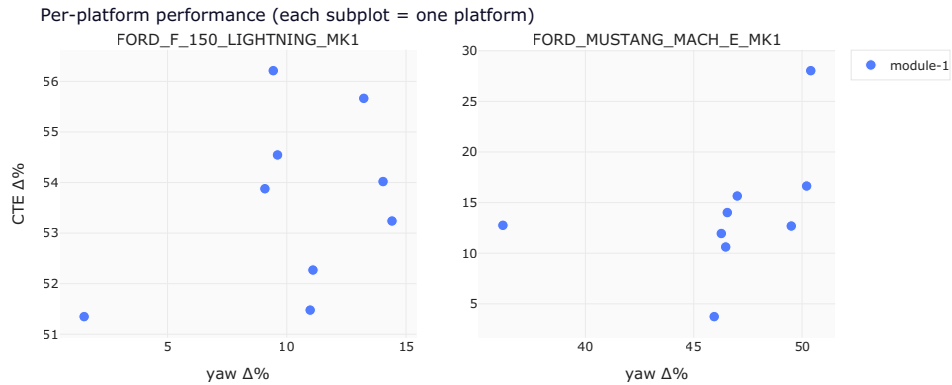
Performance grouped by which module shipped the agent.
Mean $\Delta\%$ over only the agents that reconstructed successfully;
failures shown separately.



family	n ok / total	yaw $\Delta\%$ (mean $\pm \sigma$)	CTE $\Delta\%$ (mean $\pm \sigma$)
module-1	9/10	+30.2% $\pm$ 3.6%	+24.4% $\pm$ 4.4%

• 3. Per-platform breakdown

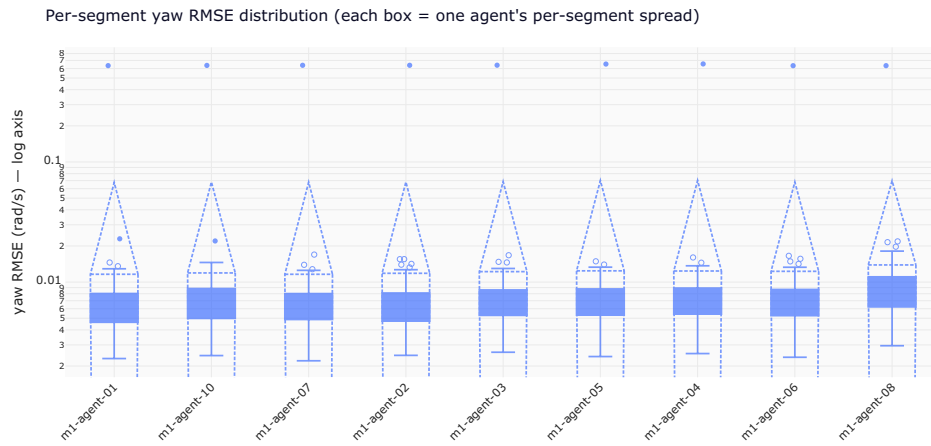
Same cohort, but split by the vehicle platform the agent was evaluated on. Agents that win one platform but lose another have a platform-specific calibration; agents whose dots cluster together generalise.



platform	agents	yaw Δ% (mean ± σ)
FORD_F_150_LIGHTNING_MK1	9	+10.4% ± 3.7%
FORD_MUSTANG_MACH_E_MK1	9	+46.5% ± 4.0%

- 4. Per-segment yaw RMSE distribution

Each box summarises one agent's per-segment yaw RMSE — pooled RMSE can hide that an agent does great on most segments but pathologically badly on a few. The y-axis is logarithmic to compress outliers.



5. Per-agent canonical scorecard

The full table. Failed reconstructions are surfaced with their reason — no fake zeros.

agent	family	status	yaw V
m1-agent-01	module-1	ok	0.01456
m1-agent-02	module-1	ok	0.01456
m1-agent-03	module-1	ok	0.01456
	module-1	ok	0.01456

agent	family	status	yaw V0
m1-agent-04			
m1-agent-05	module-1	ok	0.01456
m1-agent-06	module-1	ok	0.01456
m1-agent-07	module-1	ok	0.01456
m1-agent-08	module-1	ok	0.01456
m1-agent-09	module-1	missing_manifest_json	—
m1-agent-10	module-1	ok	0.01456

• 6. Calibration cards — fitted coefficients

Where the cohort converges on similar physics, and where it forks. Each tile is one agent's `coeffs.json` as shipped.



m1-agent-02

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.4760851016953556,
    "K_us": 0.0026722138032651867,
    "bias_rad": 0.0002686192499419401
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.809760457416018,
    "K_us": 0.0038412171248315657,
    "bias_rad": 0.001307379955568698
  },
  "HYUNDAI_IONIQ_5": {
    "L": 3.096229822470566,
    "K_us": 0.003695564361835472,
    "bias_rad": -0.0002960976826300522
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K_us": 0.0,
    "bias_rad": 0.0
  }
}
```

m1-agent-0

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "a": 0.97,
    "b": -0.0,
    "K": 0.00
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.98,
    "a": 1.19,
    "b": 3.4e-05,
    "K": 0.00
  },
  "HYUNDAI_IONIQ_5": {
    "L": 3.0,
    "a": 0.97,
    "b": 0.00,
    "K": 0.00
  },
  "TESLA_MODEL_3": {
    "L": 2.87,
    "a": 1.01,
    "b": 5e-05,
    "K": 6.7e-05
  }
}
```

m1-agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a": -0.004329037631466362,
    "b": 0.9177897626357783,
    "c": -0.00044178131403551616
  },
  "FORD_MUSTANG_MACH_E_MK1": {
```

m1-agent-0

```
{
  "model": "V",
  "platforms": "TESLA_MODEL_3",
  "L": 2.87,
  "b": 0.0,
  "Ku": 0.0
```

```
    "a": 0.0002206757945255914,  
    "b": 1.1513559927303039,  
    "c": -0.0005583363495612761  
  },  
  "HYUNDAI_IONIQ_5": {  
    "a": 0.0020222989265629857,  
    "b": 0.9120045284217976,  
    "c": -0.00048449411444108955  
  },  
  "TESLA_MODEL_3": {  
    "a": 0.0002206757945255914,  
    "b": 1.1513559927303039,  
    "c": -0.0005583363495612761  
  },  
  "_default": {  
    "a": 0.0,  
    "b": 1.0,  
    "c": 0.0  
  }  
}
```

```
    "k": 0.0,  
    "fit_score": 0.0  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "L": 2.984,  
    "b": -0.0,  
    "Ku": 0.0,  
    "k": 1.0,  
    "fit_score": 0.0  
  },  
  "FORD_F_150": {  
    "L": 3.0,  
    "b": 0.0,  
    "Ku": 0.0,  
    "k": 0.0,  
    "fit_score": 0.0  
  },  
  "HYUNDAI_IONIQ_5": {  
    "L": 3.0,  
    "b": -0.0,  
    "Ku": 0.0,  
    "k": 0.0,  
    "fit_score": 0.0  
  }  
},  
  "default": {  
    "L": 3.0,  
    "b": 0.0,  
    "Ku": 0.0,  
    "k": 1.0  
  }  
}
```

m1-agent-06

```
{  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "L": 2.984,  
    "b": -0.0,  
    "Ku": 0.0,  
    "k": 1.0,  
    "fit_score": 0.0  
  },  
  "FORD_F_150": {  
    "L": 3.0,  
    "b": 0.0,  
    "Ku": 0.0,  
    "k": 0.0,  
    "fit_score": 0.0  
  },  
  "HYUNDAI_IONIQ_5": {  
    "L": 3.0,  
    "b": -0.0,  
    "Ku": 0.0,  
    "k": 0.0,  
    "fit_score": 0.0  
  }  
}
```

m1-agent-0

```
{  
  "TESLA_MODEL_3": {  
    "L": 2.87,  
    "b": 1.0,  
    "Ku": 0.0,  
    "k": 0.0,  
    "fit_score": 0.0  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "L": 2.984,  
    "b": -0.0,  
    "Ku": 0.0,  
    "k": 1.0,  
    "fit_score": 0.0  
  },  
  "FORD_F_150": {  
    "L": 3.0,  
    "b": 0.0,  
    "Ku": 0.0,  
    "k": 0.0,  
    "fit_score": 0.0  
  }  
}
```

```

    "best": "M2_understeer",
    "m1": {
      "a": 1.0870612234849137
    },
    "m2": {
      "L_eff": 2.4763030468069056,
      "K_us": 0.0024395167795073017
    },
    "m3": {
      "a": 0.9751496691433119,
      "b": 0.23839497273290036,
      "c": 0.0004082870644824788
    },
    "val_rmse": {
      "V0": 0.0267999381028469,
      "M1": 0.027439581036525294,
      "M2": 0.026420879339395156,
      "M3": 0.031903168689021696
    },
    "train_rmse": {
      "V0": 0.012685628695945105,
      "M1": 0.011389013310887795,
      "M2": 0.008745921115146504,
      "M3": 0.010386863485176999
    }
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "best": "M2_understeer",
    "m1": {
      "a": 0.8569324797501493
    },
    "m2": {
      "L_eff": 3.882198880475702,
      "K_us": 0.003439279945738158
    },
    "m3": {
      "a": 0.8772240685582882,
      "b": -0.031353830652592465,
      "c": -0.0036354381262862774
    },
    "val_rmse": {
      "V0": 0.01591104278075153,

```

```

      "K": 5.97
      "gain": 1
      "tau": 0.
      "K_us_onl
      "gain_us_
      "variant"
    },
    "FORD_MUSTANG_MK1": {
      "L": 2.98
      "K": 0.00
      "gain": 1
      "tau": 0.
      "K_us_onl
      "gain_us_
      "variant"
    },
    "FORD_F_150_LIGHTNING_MK1": {
      "L": 3.7,
      "K": 0.00
      "gain": 0
      "tau": 0.
      "K_us_onl
      "gain_us_
      "variant"
    },
    "HYUNDAI_IONIQ_MK1": {
      "L": 3.0,
      "K": 0.00
      "gain": 0
      "tau": 0.
      "K_us_onl
      "gain_us_
      "variant"
    }
  }
}

```

```
"M1": 0.011141896721881754,
"M2": 0.0073696198273352906,
"M3": 0.0104650790814036
},
"train_rmse": {
  "V0": 0.02018452928972004,
```

m1-agent-08

```
{
  "HYUNDAI_IONIQ_5": {
    "L": 3.0,
    "K": 0.0033895612010179177,
    "s": 0.9668278221152291,
    "off": 0.0005045046778856075,
    "lag": 3,
    "rmse": 0.00888482422851739
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "K": 0.002625595512351926,
    "s": 1.1771921499482927,
    "off": 3.9923690213217876e-05,
    "lag": 4,
    "rmse": 0.01315660341959157
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "K": 0.0035388455508032707,
    "s": 0.9569361755429138,
    "off": -0.0011908292677060655,
    "lag": 3,
    "rmse": 0.011832340064259452
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K": 0.0033895612010179177,
```

m1-agent-1

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L_eff": 3.0,
    "K_us": 0.0033895612010179177,
    "d0": -4.0,
    "tau_yaw": 0.01,
    "tau_steer": 0.01
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.7,
    "K_us": 0.002625595512351926,
    "d0": 0.0,
    "tau_yaw": 0.01,
    "tau_steer": 0.01
  },
  "HYUNDAI_IONIQ_5": {
    "L_eff": 3.0,
    "K_us": 0.0033895612010179177,
    "d0": -0.0,
    "tau_yaw": 0.01,
    "tau_steer": 0.01
  },
  "TESLA_MODEL_3": {
    "_note": "passthro"
  }
}
```

```
    "s": 1.0,  
    "off": 0.0,  
    "lag": 3,  
    "rmse": null  
  }  
}
```

- **6b. Operating-contract audit**

The grader strips `sim_df` to an allowlist before calling each agent's `predict()`, so an agent cannot read truth at inference time. This section reports source-level intent — which agents *would have* read truth columns if the grader hadn't intervened. A clean audit is its own quality signal.

Clean cohort: no agent referenced truth-derived column names inside their `predict()` body.

- **7. Reconstruction quality (substrate signal)**

How many agents shipped the right artefacts to be canonically gradable. Failures here are a substrate or contract problem, not a model problem.

format check	pass	fail
agent_folder_exists	10	0

format check	pass	fail
has_manifest_json	9	1
manifest_parsable	9	1
manifest_declares_predict_callable	9	1
manifest_declares_platform_support	9	1
has_predict_py	9	1
has_coeffs_json	8	2
has_report	0	10

Failure reasons across the cohort:

- missing_manifest_json — 1 agent(s)

8. Worst-of-cohort

Lowest yaw $\Delta\%$

- m1-agent-08
(+20.9%)
- m1-agent-05
(+29.6%)

Lowest CTE $\Delta\%$

- m1-agent-10
(+16.5%)
- m1-agent-05
(+21.8%)

- m1-agent-04
(+29.7%)

- m1-agent-08
(+23.0%)

Generated by grade-cohort-reports at 2026-05-31 21:46 · idea idea-01-lateral-attribution